Polar Form, De Moivre's Theorem & Roots of Unity

Section 1: Polar and Euler Forms

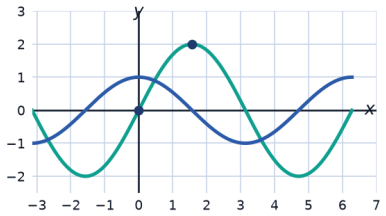
1 Express $z = -2 - 2\sqrt{3}i$ in Euler's exponential form $re^{i\theta}$ with $\theta \in (-\pi, \pi]$.

a. $4e^{-i2\pi/3}$

b. $4e^{i4\pi/3}$

c. $2e^{-i2\pi/3}$

d. $4e^{-i\pi/3}$



2 Convert $z = 6e^{i3\pi/4}$ into exact Cartesian form $x + iy$.

a. $-3\sqrt{2} + 3\sqrt{2}i$

b. $3\sqrt{2} - 3\sqrt{2}i$

c. $-3 + 3\sqrt{3}i$

d. $-3\sqrt{2} - 3\sqrt{2}i$

3 If $z_1 = 3e^{i\pi/3}$ and $z_2 = 2e^{i\pi/6}$, evaluate the quotient $\frac{z_1}{z_2}$ in exponential form.

a. $\frac{3}{2}e^{i\pi/6}$

b. $\frac{3}{2}e^{i\pi/2}$

c. $6e^{i\pi/6}$

d. $1.5e^{-i\pi/6}$

Section 2: De Moivre's Theorem for Powers

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4 Use De Moivre's Theorem to evaluate $(1 + i\sqrt{3})^6$.

a. 64

b. -64

c. $64i$

d. $-64i$

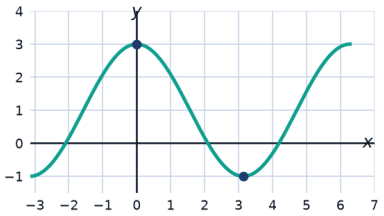
5 Evaluate $(\cos(\pi/5) + i\sin(\pi/5))^{10}$.

a. 1

b. -1

c. i

d. $-i$



6 Express $\cos(3\theta)$ in terms of $\cos\theta$ using De Moivre's Theorem $(\cos\theta + i\sin\theta)^3$.

a. $4\cos^3\theta - 3\cos\theta$

b. $3\cos^3\theta - 4\cos\theta$

c. $\cos^3\theta - 3\cos\theta$

d. $4\cos^3\theta + 3\cos\theta$

Section 3: Roots of Complex Numbers

7 Find all cube roots of $8i$ in polar form.

a. $2e^{i\pi/6}, 2e^{i5\pi/6}, 2e^{-i\pi/2}$

b. $2e^{i\pi/3}, 2e^{i\pi}, 2e^{i5\pi/3}$

c. $8e^{i\pi/6}, 8e^{i5\pi/6}, 8e^{-i\pi/2}$

d. $2, -1 + i\sqrt{3}, -1 - i\sqrt{3}$

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8 How are the n distinct nth roots of any complex number $z = re^{i\theta}$ distributed on the Argand plane?

a. They lie on a circle of radius $r^{1/n}$ centered at the origin, evenly spaced by $\frac{2\pi}{n}$ radians

b. They lie along a straight line through the origin

c. They form an ellipse with eccentricity $1/n$

d. They cluster exclusively in Quadrant I

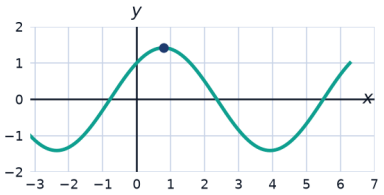
9 What is the sum of all n distinct nth roots of any complex number z ($n \geq 2$)?

a. 0

b. 1

c. n

d. z



Section 4: Roots of Unity & Factorization

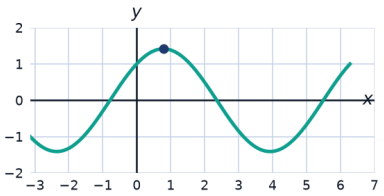
10 If $\omega = e^{i2\pi/3}$ is a primitive complex cube root of unity, what is the value of $1 + \omega + \omega^2$?

a. 0

b. 1

c. -1

d. 3



11 Factor $z^4 + 1$ into two real quadratic factors.

a. $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)$

b. $(z^2 + 1)^2$

c. $(z^2 - 1)(z^2 + 1)$

d. $(z^2 - 2z + 1)(z^2 + 2z + 1)$

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12 The four 4th roots of unity ($z^4 = 1$) form the vertices of which geometric shape on the Argand plane?

a. A square inscribed in the unit circle with vertices at $\pm 1, \pm i$

b. An equilateral triangle

c. A regular hexagon

d. A rectangle of perimeter 4

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